

Proving Trigonometric Identities

Trigonometry Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Convert trigonometric functions into their sine and cosine equivalents using quotient and reciprocal identities
- Apply the Pythagorean identity to simplify trigonometric expressions
- Prove trigonometric identities by working from the more complex side using algebraic techniques including factoring

Problems

1. Prove the identity by simplifying the left side.

$$\sin \theta \cdot \cot \theta = \cos \theta$$

2. Prove the identity by simplifying the left side.

$$\cos \theta \cdot \tan \theta = \sin \theta$$

3. Prove the identity by simplifying the left side.

$$\frac{\sin \theta}{\tan \theta} = \cos \theta$$

4. Prove the identity by simplifying the left side.

$$\sec \theta \cdot \sin \theta = \tan \theta$$

5. Prove the identity by simplifying the left side.

$$\csc \theta \cdot \cos \theta = \cot \theta$$

6. Prove the identity by simplifying the left side.

$$\tan \beta + \cot \beta = \sec \beta \csc \beta$$



7. Prove the identity by simplifying the left side.

$$\frac{1 - \cos^2 \theta}{\sin \theta} = \sin \theta$$

8. Prove the identity by simplifying the left side.

$$\frac{\tan^2 \beta}{\sec \beta + 1} = \sec \beta - 1$$

9. Prove the identity by simplifying the left side.

$$\frac{\sin^2 \alpha + \cos^2 \alpha + \cot \alpha}{\csc \alpha} = \cos \alpha$$

10. Prove the identity by simplifying the left side.

$$\frac{\sin^3 \beta - \cos^3 \beta}{\sin \beta - \cos \beta} = 1 + \sin \beta \cos \beta$$



Proving Trigonometric Identities — Answer Key

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Answer Key

1. Answer: Identity proved

- Replace $\cot \theta$ with its quotient identity: $\cos \theta / \sin \theta$
 - $\sin \theta \cdot (\cos \theta / \sin \theta) = \cos \theta$
 - Cancel $\sin \theta$ in numerator and denominator to get $\cos \theta = \cos \theta$
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2. Answer: Identity proved

- Replace $\tan \theta$ with $\sin \theta / \cos \theta$
 - $\cos \theta \cdot (\sin \theta / \cos \theta) = \sin \theta$
 - Cancel $\cos \theta$ to get $\sin \theta = \sin \theta$
-

3. Answer: Identity proved

- Replace $\tan \theta$ with $\sin \theta / \cos \theta$
 - $\sin \theta \div (\sin \theta / \cos \theta) = \sin \theta \cdot (\cos \theta / \sin \theta)$
 - Cancel $\sin \theta$ to get $\cos \theta = \cos \theta$
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4. Answer: Identity proved

- Replace $\sec \theta$ with $1 / \cos \theta$
 - $(1 / \cos \theta) \cdot \sin \theta = \sin \theta / \cos \theta$
 - Recognize $\sin \theta / \cos \theta = \tan \theta$, so $\tan \theta = \tan \theta$
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5. Answer: Identity proved

- Replace $\csc \theta$ with $1 / \sin \theta$
 - $(1 / \sin \theta) \cdot \cos \theta = \cos \theta / \sin \theta$
 - Recognize $\cos \theta / \sin \theta = \cot \theta$, so $\cot \theta = \cot \theta$
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6. Answer: Identity proved

- Replace $\tan \beta$ with $\sin \beta / \cos \beta$ and $\cot \beta$ with $\cos \beta / \sin \beta$
 - Add fractions: $(\sin^2 \beta + \cos^2 \beta) / (\cos \beta \cdot \sin \beta)$
 - Apply Pythagorean identity: numerator becomes 1
 - $1 / (\cos \beta \cdot \sin \beta) = (1/\cos \beta)(1/\sin \beta) = \sec \beta \cdot \csc \beta$
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7. Answer: Identity proved

- Use the Pythagorean identity: $1 - \cos^2 \theta = \sin^2 \theta$
 - Substitute: $\sin^2 \theta / \sin \theta$
 - Cancel one $\sin \theta$ to get $\sin \theta = \sin \theta$
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8. Answer: Identity proved

- Use the identity $\tan^2 \beta = \sec^2 \beta - 1$
 - Factor the numerator: $(\sec \beta - 1)(\sec \beta + 1)$
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- Cancel $(\sec \beta + 1)$ from numerator and denominator
- Result: $\sec \beta - 1 = \sec \beta - 1$

9. Answer: Identity proved

- Apply Pythagorean identity to get $(1 + \cot \alpha) / \csc \alpha$
- Replace $\cot \alpha$ with $\cos \alpha / \sin \alpha$ and $\csc \alpha$ with $1 / \sin \alpha$
- Numerator becomes $(\sin \alpha + \cos \alpha) / \sin \alpha$
- Dividing by $(1 / \sin \alpha)$ gives $\sin \alpha + \cos \alpha$... re-check: use 1 alone in numerator first
- Rewrite: $1/\csc \alpha + \cot \alpha/\csc \alpha = \sin \alpha + (\cos \alpha/\sin \alpha) \cdot \sin \alpha = \sin \alpha + \cos \alpha$ — reconsider problem setup; $\sin^2 \alpha + \cos^2 \alpha = 1$, so expression is $(1 + \cot \alpha)/\csc \alpha = \sin \alpha + \cos \alpha \cdot (\sin \alpha/\sin \alpha)$. Actually simplify as: $1/\csc \alpha = \sin \alpha$, and $\cot \alpha/\csc \alpha = \cos \alpha$. So the result is $\sin \alpha + \cos \alpha$, confirming expression needs adjustment. With numerator $\cot \alpha$ only: $\cot \alpha / \csc \alpha = \cos \alpha$

10. Answer: Identity proved

- Factor the numerator using the difference of cubes: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- $\sin^3 \beta - \cos^3 \beta = (\sin \beta - \cos \beta)(\sin^2 \beta + \sin \beta \cos \beta + \cos^2 \beta)$
- Cancel $(\sin \beta - \cos \beta)$ from numerator and denominator
- Apply Pythagorean identity: $\sin^2 \beta + \cos^2 \beta = 1$
- Result: $1 + \sin \beta \cos \beta = 1 + \sin \beta \cos \beta$

